

Assignment 3

(Points 5 – Group/Solo based)

Assigned: 17th March

Deadline: 31st March

CS 2123: DATA STRUCTURES

Assignment 3

- Work Submission:
 - Students should submit a zip file (Assign3.zip) containing all files (i.e. 1 PDF and 2 code files) on the Canvas.
 - 1 PDF = Huffman Encoding (details about PDF are on the next slide)
 - 2 Code files = Huffman variable bit and Heap tree
 - If you are working as a group, group members names and ABC123 should be printed out with the intended output of the codes
 - Each member will submit the same zip file from their Canvas
 - If working solo, only your name and ABC123 should be printed out with intended output of the codes
 - Kindly **do not email** the assignment; graders can only grade your assignment on Canvas.
 - Late work will be penalized with half points deduction (As mentioned in the outline). The only exception will be documented, extenuating circumstances, which must be communicated prior to the due date or submission of the work.
 - **Note:** Canvas uses 11:59:00 p.m. as the cutoff time, not 11:59:59 p.m.
- **Recommendation:** Don't submit work so close to the deadline that you have to worry about bandwidth issues, Canvas issues, or discrepancies in your system clock and the Canvas clock. These will not be considered extenuating circumstances.
- **Plagiarism:** Plagiarism will result in zero points
- Grader will use fox server to check the codes

Part –I : Huffman Encoding (Points 2)

- In this assignment you must create a Huffman Encoding (Variable) Table and Tree.
 - *Note: Code the Huffman variable bit*

Huffman Code/bit **representation of AAAAAABBBCCDDEEEFFFFF (characters)**. This is the fix string; students do not need to program for user input.

- Variable bit table representation (Your code will generate the coded table as output Fig A, points: 1)
- PDF file containing the complete table of characters, bit codes, frequencies, total bits (i.e., Table A) and its tree representation. Students cannot submit hand drawn tree. (point: 1)

```
s:000
g:0010
o:0011
i:010
a:011
b:10
p:110
t:11100
e:11101
n:1111
```

Fig A

Note: Fig A & Table A are examples, not encoding characters for the assignment

Table A

Approach (example)				
Char	Freq	Code	Total	
B	5	10	10	
I	2	010	6	
G	1	0010	4	
p	3	110	9	
O	1	0011	4	
T	1	11100	5	
E	1	11101	5	
S	2	000	6	
A	3	011	9	
N	2	1111	8	
Total Bits to transmit			66	

Part –II : Heap Tree (Points 3 – Group/solo based)

- The code will ask user for 7 integers (1 point)
 - User will press enter to provide integer input values.
 - No Duplicate Value will be entered by the user (by default)
- The code will ask user for Min Heap or Max Heap (After 7 integers are entered) (0.5 points)
 - Press 1 for Min Heap
 - Press 2 for Max Heap
- Then It will out put the desired Heap tree as follow: (1.5 points)
 - If user selected Max Heap Fig B
 - If use selected Min Heap Fig C
- Submission:
 - Only Code .java file is to be submitted

Extracted Max: 13
Extracted Max: 10
Extracted Max: 6
Extracted Max: 5
Extracted Max: 4
Extracted Max: 3
Extracted Max: 1

Fig B

Extracted Min: 1
Extracted Min: 3
Extracted Min: 4
Extracted Min: 5
Extracted Min: 6
Extracted Min: 10
Extracted Min: 13

Fig C